Name: _____

Class: _____Date:_____

LESSON 9. DIVISIBILITY RULES

2.12. Which of the following numbers are divisible by both 2 and 3?

4 016; 4 014; 5 212; 5 214.

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2.13. Which of the following numbers are divisible by both 3 and 5?

2 035; 3 501; 3 105; 2 045.

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2.14. Which of the following numbers are divisible by both 2 and 9?

4 518; 4 527; 4 506; 4 510.

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2.15. Without calculating, decide whether each sum (difference) below is divisible by 2.

a) 2 034 + 2 046; b) $2\,025^3 - 2\,024^3$.

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2.16. Without calculating, decide whether each sum (difference) below is divisible by 5.

a) 2 035 + 2 041; b) $2\,030^4 - 2\,015^3$.

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2.17. Class 6A wants to form a dance group that can be split evenly into groups of 2 people or into groups of 9 people. What is the smallest number of people the dance group needs?

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2.18. Use the four digits 0; 4; 7; 5 to make four-digit numbers, using each digit exactly once, such that:

a) the numbers are divisible by 2; b) the numbers are divisible by 5;

c) the numbers are divisible by both 2 and 5.

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2.19. Given the number n = $\overline{417ab}$. Replace a and b with suitable digits so that n is divisible by both 5 and 9.

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2.20. To get ready for the new school year, Mai's mother bought her some pens and some notebooks for a total of 160 thousand dong. Each pen costs 19 thousand dong and each notebook costs 5 thousand dong. How many pens and how many notebooks did she buy?

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2.21. Is each sum below divisible by 3? Why?

a) $A = 10^{15} + 5$;

b) $B = 10^{15} + 4$.

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2.22. Is each sum below divisible by 9? Why?

a) $A = 10^{20} + 8$;

b) $B = 10^{20} + 6$.

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